5.1.2 Solid, liquid and gas

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	solids, liquids and gases in terms of the spacing, ordering and motion of atoms or molecules	
(b)	simple kinetic model for solids, liquids and gases	
(c)	Brownian motion in terms of the kinetic model of matter and a simple demonstration using smoke particles suspended in air	
(d)	internal energy as the sum of the random distribution of kinetic and potential energies associated with the molecules of a system	

(e) absolute zero (0 K) as the lowest limit for temperature; the temperature at which a substance has minimum internal energy

(f) increase in the internal energy of a body as its temperature rises

(g) changes in the internal energy of a substance during change of phase; constant temperature during change of phase.